Adaptive Sinhala Sign Language Learning System

25-26J-223

Project Proposal Report

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Declaration

I declare that the work presented here is original and that we have not plagiarized in any way, shape, or form any portion of this proposal into another work that has already been submitted for a degree or diploma in any other university or higher educational institution without acknowledgement.

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The supervisor and co-supervisor should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor Date 29/8/2025

Signature of the co-supervisor Date 28/08/2025

Abstract

Sinhala Sign Language (SSL) is the primary mode of communication for the Deaf community in Sri Lanka. However, learning SSL remains challenging due to the lack of interactive, adaptive, and learner-centered digital platforms. Existing sign learning systems are mostly static, offering fixed lessons without considering individual learning speed, mistakes, or progress. As a result, learners often struggle to master correct hand gestures, transitions, and consistency, leading to reduced motivation and ineffective learning outcomes. This research proposes an Adaptive Sinhala Sign Language Learning System that personalizes the learning experience using Artificial Intelligence and reinforcement learning principles. The system focuses on recognizing learner performance through hand gesture analysis and dynamically adjusting learning paths based on accuracy, repetition needs, and progression readiness. Instead of advancing learners uniformly, the platform adapts lesson difficulty, repetition frequency, and feedback intensity according to individual performance levels. The proposed system initially targets static and dynamic Sinhala sign letters, with future extensibility to vowels, numbers, and word-level signs based on user feedback. Real-time corrective feedback, visual hints, and performance scoring guide learners toward correct signing techniques. By providing a personalized and adaptive learning journey, the system enhances engagement, accuracy, and retention. This research contributes to inclusive education by delivering a culturally relevant, learner-adaptive SSL learning platform, addressing a critical gap in current sign language education technologies in Sri Lanka.

Keywords: Sinhala Sign Language, Adaptive Learning, Reinforcement Learning, Gesture Recognition, Assistive Technology

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Sinhala Sign Language	SSL
Reinforcement Learning	RL
Artificial Intelligence	AI

1. INTRODUCTION

The deaf and hearing-impaired face immense difficulties in learning sign language proficiency because of the lack of availability of learning materials in an interactive platform. Although the Sinhala Sign Language is the important medium of communication in Sri Lanka for the deaf community, most materials available for learning the language are in the form of static images or recorded videos or learning classes with the instructor. [1],[2].

To successfully develop this project, strong expertise in several areas is required, including:

1. **Sign Gesture Recognition:** The ability to accurately detect and evaluate Sinhala sign gestures using computer vision and deep learning techniques such as Convolutional Neural Networks and hand landmark detection models.
2. **Adaptive Learning and Reinforcement Learning:** Applying reinforcement learning principles to dynamically adjust lesson progression, repetition frequency, and difficulty levels based on learner performance and accuracy.
3. **Human-Centered Learning Systems:** Designing learner-focused feedback mechanisms that provide clear visual guidance, performance indicators, and corrective suggestions to support continuous improvement.

At present, the existing learning models for Sinhala sign language are primarily based on content display without any performance assessment or adaptation [8]. The learners are not even aware of the mistakes they make, and they are progressed to the next level without Mastering the previous signs efficiently. Even the real-time learning models are not implemented in the SSL learning platforms for the Sinhala sign language, though it is a low-resource language.

Several studies have demonstrated the effectiveness of adaptive learning systems and AI-driven feedback in skill-based education [12]. Similarly, research in gesture recognition has shown promising accuracy in detecting hand signs. However, the integration of these technologies into a complete, learner-adaptive Sinhala sign learning platform, especially one suitable for mobile deployment remains largely unexplored[5].

This project addresses these gaps by proposing an Adaptive Sinhala Sign Language Learning System that personalizes the learning experience through real-time gesture recognition and reinforcement learning-based progression control. Unlike traditional static learning applications, the proposed system continuously evaluates learner performance, provides corrective feedback, and adapts lesson flow to individual needs. By introducing an intelligent, learner-centered approach to SSL education, this research aims to improve learning accuracy, engagement, and long-term skill retention, contributing to more inclusive and effective sign language education in Sri Lanka.

1.1 Background and Literature Survey

It is important to note that sign language learning forms a critical ingredient in the improvement of accessible communication for deaf and hard-of-hearing individuals, as well as for hearing individuals looking to learn sign language for inclusive interactions. Conventionally, sign language education has been carried out using instructor-led sessions, printed materials, or demonstrations through recorded videos[4]. Recently, there has been development in the areas of mobile technologies and computer vision; because of this, various sign language learning applications have emerged. However, despite these technological advances, a lot of the existing systems still revert to static learning methodologies that do not effectively address individual learner needs.

Most current sign language learning applications present signs using pre-recorded videos or images and require learners to imitate the demonstrated gestures. These systems typically follow a fixed lesson structure, where all users progress through the same sequence of signs regardless of their learning speed, accuracy, or error frequency. Such static learning models fail to accommodate the diverse learning abilities of users and do not support mastery-based progression, which is especially important for acquiring precise motor skills such as hand shapes, finger positioning, and orientation in sign language.

Sign language research and development activity has been predominantly geared towards sign recognition and sign to text or sign to speech translation systems and not learning systems. Several research publications have used computer vision approaches and deep learning algorithms, specifically CNNs, to develop hand gesture recognition systems [8][9]. The idea is to extract spatial features related to hand contours, fingertips, palm orientations, and movement paths from images and videos. While this area of research and development has produced very encouraging results related to high recognition rates, its focus is not learning but recognition or translation.

One of the critical limitations found in existing solutions for learning sign language through computers is the absence of corrective feedback during runtime. In prevalent solutions, the user only gets feedback in the form of correct or incorrect signals for the entered sign language gesture. The solutions do not provide any explanation for which part of the hand gesture signal is incorrect, be it the fingers or the palm[11][12].

positioning, hand orientation, or movement execution. As a result, learners may unknowingly repeat incorrect gestures, reinforcing improper signing habits. Educational research emphasizes that timely and informative feedback is essential for effective skill acquisition, particularly in motor learning tasks [7].

Furthermore, progression mechanisms in current platforms are often time-based or manually controlled, rather than driven by user performance. Learners may advance to new signs without fully mastering previous ones, which can lead to confusion and reduced retention. This approach contrasts with adaptive learning principles, where content difficulty and progression are dynamically adjusted based on learner performance metrics such as accuracy, consistency, and response time [3].

Another major shortcoming in existing sign language learning systems is the absence of adaptive learning mechanisms. Adaptive learning systems aim to personalize the learning experience by analyzing user performance and modifying instructional strategies accordingly. Reinforcement Learning, a branch of machine learning, has been successfully applied in various educational domains to model learner behavior and optimize learning paths. However, the application of reinforcement learning in sign language education remains largely unexplored. Most existing platforms do not adjust repetition frequency, difficulty level, or feedback style based on individual learner progress.

This limitation is particularly evident in the context of low-resource sign languages such as Sinhala Sign Language, where research, datasets, and learning tools are limited. While several studies focus on widely used sign languages such as American Sign Language (ASL), comparatively little attention has been given to developing intelligent learning systems for regional or low-resource sign languages. The lack of adaptive and personalized learning tools further widens the accessibility gap for learners of Sinhala Sign Language [1].

The limitations identified in existing literature namely static learning structures, lack of adaptive progression, absence of corrective feedback, and limited focus on learning-centered design highlight a clear research gap. Addressing this gap requires an intelligent learning system that evaluates user performance in real time, provides personalized feedback, and dynamically adjusts learning paths based on learner behavior. The proposed Adaptive Sign Learning System seeks to address these challenges by integrating real-time hand gesture recognition with reinforcement learning-based personalization, thereby offering a more effective, scalable, and learner-centric approach to sign language learning.

1.2 Research Gap

Although sign language learning systems have gained increasing attention with the advancement of computer vision and machine learning technologies, significant research gaps remain, particularly in the context of Sinhala Sign Language education. Most existing sign language learning applications adopt static instructional approaches, where learners are guided through a predefined and fixed lesson structure. These systems do not account for individual learner differences such as learning pace, accuracy levels, repetition needs, or error patterns. As a result, learners are forced to follow the same learning path regardless of their proficiency, leading to ineffective skill acquisition and reduced learner engagement.

A critical gap identified in current sign language learning systems is the absence of meaningful feedback mechanisms. Many existing platforms provide sign demonstrations through images or videos but do not validate whether the learner performs the sign correctly. In some cases, systems simply indicate a correct or incorrect response without explaining the specific nature of the error. This lack of corrective feedback prevents learners from understanding mistakes related to finger placement, hand orientation, or movement accuracy. Without detailed guidance, learners may repeatedly practice incorrect gestures, which negatively impacts learning quality and long-term retention.

Furthermore, most sign learning systems lack personalized lesson paths. Progression is often time-based or manually controlled rather than performance driven. Learners who struggle with certain signs are not provided with additional practice or simplified content, while faster learners are required to complete unnecessary repetitions. Educational research consistently highlights that adaptive learning systems significantly improve outcomes by tailoring instruction based on learner performance. However, such adaptive strategies are largely absent in sign language learning platforms, particularly those targeting low-resource sign languages such as Sinhala Sign Language.

Another important research gap relates to the limited application of reinforcement learning in sign language education. While reinforcement learning has been successfully applied in intelligent tutoring systems and adaptive educational platforms, its integration into sign language learning remains minimal. Existing SSL learning tools do not utilize learner performance metrics such as success rate, repetition count, or consistency to dynamically adjust lesson difficulty, feedback strategies, or learning progression. This lack of intelligence-driven adaptation limits the system's ability to support individualized learning experiences.

In terms of content coverage, current SSL learning systems often provide incomplete linguistic scope. Many applications focus primarily on a small subset of static letters and fail to comprehensively cover Sinhala alphabets, numerical signs, and commonly used words. This restricted content limits learners' ability to achieve functional communication skills required for real-world interactions. Additionally, most platforms do not structure content into progressive learning levels (basic, intermediate, advanced) that align with learner competency, further reducing learning motivation and goal-oriented progression.

Another notable limitation is that existing systems prioritize recognition accuracy over educational effectiveness. Many research efforts are centered on sign recognition or sign-to-text translation rather than learner-centric system design. As a result, these systems function more as recognition tools rather than interactive learning environments. They do not support mastery-based progression, adaptive repetition, or feedback-driven learning, which are essential components of effective skill-based education.

Collectively, these limitations highlight a substantial research gap in the development of an adaptive, feedback-driven Sinhala Sign Language learning system. There is a clear need for a solution that combines real-time hand gesture recognition with intelligent personalization, enabling learners to receive corrective feedback and follow customized lesson paths based on their performance. Addressing this gap requires the integration of reinforcement learning techniques to dynamically adapt learning content, feedback intensity, and progression strategies.

The proposed Adaptive Sinhala Sign Language Learning System aims to bridge this gap by introducing a learner-centric approach that supports personalized lesson paths, real-time corrective feedback, and comprehensive coverage of Sinhala alphabets, numbers, and words. By leveraging machine learning ba

sed hand gesture recognition and reinforcement learning driven adaptation, the system transforms SSL learning from a static and passive process into an intelligent, interactive, and effective educational experience.

1.3 Research Problem

Sign language learning plays a vital role in enabling inclusive communication for deaf and hard-of-hearing individuals, as well as for hearing individuals seeking to learn sign language for social, educational, or professional purposes. Sinhala Sign Language, in particular, remains underrepresented in terms of intelligent digital learning tools, despite its importance within the Sri Lankan deaf community. Although several mobile and web-based sign language learning applications exist, most of them rely on static instructional methods that do not effectively support individualized learning or skill mastery. The primary research problem addressed in this study is the lack of an adaptive, learner-centered learning mechanism in existing Sinhala Sign Language learning systems. Current platforms typically present a fixed sequence of lessons consisting of images or videos demonstrating hand gestures. Learners are expected to imitate these signs without receiving systematic evaluation or guidance. Such systems do not assess learner performance in real time and therefore fail to adjust learning content based on user accuracy, learning speed, or repetition needs. This static approach significantly limits the effectiveness of sign language acquisition, which is inherently a motor-skill–based learning process requiring continuous practice, correction, and reinforcement.

A major challenge within this problem is the absence of real-time performance evaluation and feedback. Most existing sign learning systems either do not provide feedback at all or offer only simple correctness indicators. They do not explain why a sign is incorrect or identify specific errors related to hand shape, finger positioning, or orientation. Without meaningful corrective feedback, learners are unable to self-correct effectively and may unknowingly reinforce incorrect signing patterns. This issue is particularly problematic for beginners, who depend heavily on guidance to develop correct foundational skills.

Another critical aspect of the research problem is the lack of personalized lesson progression. Learners exhibit significant variation in learning pace, consistency, and motor coordination. However, existing systems do not adapt lesson difficulty or repetition frequency based on individual performance. Learners who struggle with certain signs are often forced to progress prematurely, while faster learners are required to follow unnecessarily repetitive content. This mismatch between learner needs and instructional delivery leads to frustration, reduced engagement, and inefficient learning outcomes.

Furthermore, current Sinhala Sign Language learning applications often provide limited content coverage, focusing mainly on a small subset of static letters. Comprehensive learning of Sinhala alphabets, numerical signs, and commonly used words is rarely supported within a unified learning framework. This limitation prevents learners from developing functional sign language proficiency applicable to real-world communication scenarios. Additionally, most platforms do not structure content into clear learning levels such as basic, intermediate, and advanced stages that align with learner competence. From a technological perspective, although computer vision and deep learning techniques have been widely used for sign gesture recognition, these technologies are primarily applied for recognition or translation purposes rather than educational support. There is a lack of systems that utilize learner performance data such as accuracy rate, repetition count, and error frequency to guide instructional decisions. Reinforcement learning, which has demonstrated success in adaptive tutoring and intelligent educational systems, has not been adequately explored in the context of sign language learning, especially for low-resource languages like Sinhala. The research problem, therefore, lies in designing an intelligent learning system that can accurately evaluate hand gesture performance, identify learner proficiency levels, and dynamically adapt lesson progression and feedback strategies. The system must operate efficiently in a mobile environment, focus specifically on hand gesture recognition (without relying on emotion detection or full-body posture), and support offline usage to ensure accessibility.

Addressing this research problem is essential to transform Sinhala Sign Language learning from a static, demonstration-based experience into an adaptive, interactive, and learner-centric process. By integrating real-time hand gesture recognition with reinforcement learning based personalization, the proposed system aims to provide meaningful feedback, customized learning paths, and structured progression across Sinhala alphabets, numbers, and words. Solving this problem will contribute to more effective SSL education, improved learner engagement, and broader accessibility to intelligent assistive learning technologies.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to design and develop an Adaptive Sinhala Sign Language Learning System that supports effective and personalized learning through real-time hand gesture recognition, performance-based evaluation, and reinforcement learning–driven lesson adaptation. The proposed system aims to overcome the limitations of existing static sign language learning platforms by dynamically adjusting the learning path based on each learner's performance. Unlike traditional systems that follow a fixed lesson sequence and provide minimal feedback, this system continuously monitors user accuracy, consistency, and improvement trends to guide learners through a personalized progression. The system focuses exclusively on hand gesture recognition for Sinhala Sign Language, ensuring simplicity, efficiency, and suitability for real-time mobile execution. By integrating artificial intelligence techniques with educational principles, the system seeks to improve learning efficiency, reduce repeated errors, and increase learner engagement. The goal is to provide a learner-centric, adaptive, and accessible mobile solution that enhances the acquisition of Sinhala sign language skills for beginners and intermediate learners.

2.2 Sub Objectives

1. One of the primary sub-objectives of this research is to develop a robust hand gesture recognition module capable of accurately identifying Sinhala sign language gestures performed by learners. This module will use computer vision and deep learning techniques to capture and analyze hand shapes, finger positions, and movement patterns through the device camera. The recognition module is designed specifically for Sinhala alphabets, numbers, and selected basic words, addressing the lack of learning tools for low-resource sign languages. By focusing only on hand gestures and excluding facial expressions or body posture, the system remains computationally efficient and suitable for real-time mobile environments. Accurate gesture recognition is essential, as it forms the foundation for evaluating learner performance and enabling adaptive learning decisions.

2. Providing real-time corrective feedback is another important objective of the proposed system. When a learner performs a sign incorrectly, the system will offer immediate guidance rather than allowing repeated mistakes. Feedback may include messages such as retry prompts, visual alignment hints, or slower

progression suggestions based on the learner's difficulty level. This objective addresses a major limitation of existing platforms, where users often receive little or no explanation for errors. By offering targeted and timely feedback, the system helps learners understand their mistakes and improve their performance more efficiently, leading to better skill retention and confidence.

3. A central sub-objective of this research is to design and integrate a RL based adaptive learning engine. The RL agent will analyze learner performance metrics, including accuracy, repetition count, and improvement trends, to decide the most suitable next learning action. Based on these observations, the system may recommend repeating the same sign, providing additional practice examples, offering visual hints, or advancing to the next lesson. This adaptive decision-making process ensures that learners progress at their own pace rather than following a rigid, time-based structure. The inclusion of reinforcement learning enables the system to continuously optimize learning strategies, making the learning experience more effective and personalized.

4. The system aims to implement a progressive, level-based learning framework that unlocks content based on learner mastery rather than fixed schedules. Beginner-level Sinhala letters will be accessible by default, while intermediate and advanced signs will be unlocked only after satisfactory performance in earlier levels. This objective ensures structured learning while maintaining flexibility. Learners who perform well can progress faster, while those who need more practice receive additional support. This mastery-driven approach aligns with educational best practices and promotes long-term learning effectiveness.

5. The final sub-objective is to design a simple, intuitive, and learner-friendly mobile interface that supports interactive sign language practice. The interface will clearly display target signs, camera previews, feedback messages, and progression indicators. A well-designed UI is essential to ensure usability, reduce cognitive load, and encourage consistent practice. This objective ensures that the technical capabilities of the system are delivered through an accessible and engaging user experience suitable for learners of varying technical backgrounds.

3. METHODOLOGY

The proposed Adaptive Sinhala Sign Language Learning System follows a learner-centric and performance-driven methodology, where user interaction, gesture recognition, and adaptive feedback are tightly integrated. The system is designed to continuously monitor learner performance and dynamically adjust lesson progression using reinforcement learning techniques. This methodology ensures that learning is personalized, efficient, and aligned with individual user capabilities.

3.1 System Architecture and Workflow

3.1.1 System Diagram

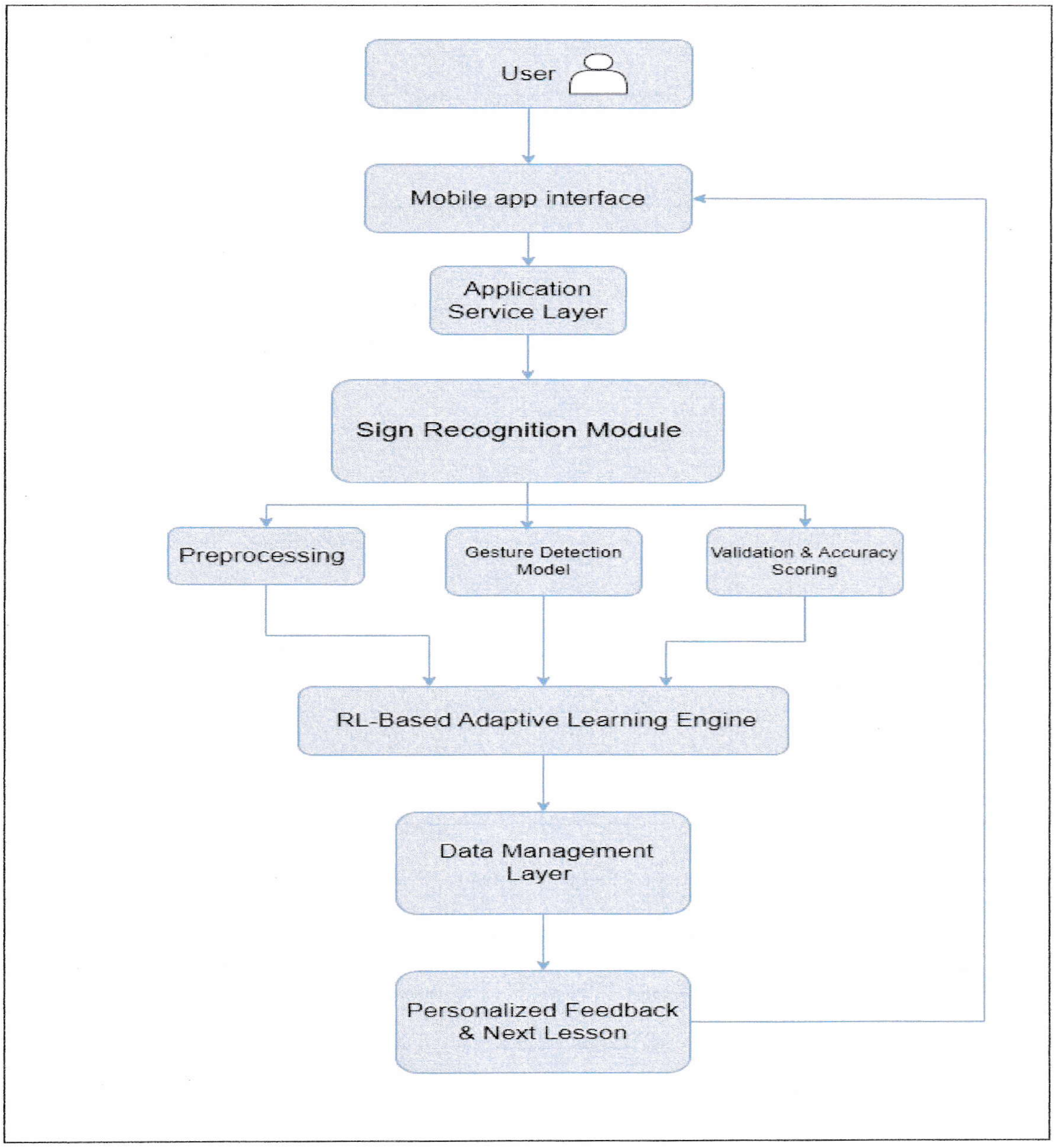

Figure 3-1 System Diagram

3.1.2 System Workflow

1. User Interaction – Learner selects a sign lesson
2. Gesture Capture – Camera captures hand movements
3. Gesture Recognition – AI model evaluates correctness
4. Performance Scoring – Accuracy score is calculated
5. Adaptive Decision Engine – Reinforcement learning agent decides next action
6. Feedback Delivery – Visual hints, repetition, or advancement

3.2 Data Requirements and Collection

The system relies on multiple data types to support gesture recognition and adaptive learning.

Sign Language Dataset:

The dataset consists of Sinhala Sign Language hand gesture images and short video samples representing static and dynamic signs. These samples cover Sinhala alphabets, numbers, and selected basic words.

Annotated Training Data:

Each gesture sample is manually labeled with its corresponding sign class to ensure accurate supervised learning. Annotation focuses on hand shape, finger positioning, and orientation.

Performance Logs:

During usage, the system collects learner interaction data such as accuracy scores, repetition frequency, and lesson progression. These logs are stored locally and used by the reinforcement learning engine to personalize future learning decisions.

3.3 Tasks and Subtasks

1. Dataset Preparation and Labeling:

Raw image and video samples are collected, cleaned, and labeled according to predefined Sinhala sign categories.

2. Model Training for Gesture Recognition:

A deep learning model is trained using the prepared dataset to recognize hand gestures with high accuracy under varying lighting and background conditions.

3. Reinforcement Learning Policy Design:

A reinforcement learning policy is developed to determine optimal learning actions based on learner performance. This includes defining states, actions, and reward mechanisms.

4. UI Development for Adaptive Feedback:

The mobile interface is designed to display lessons, camera previews, feedback messages, and progress indicators in an intuitive manner.

5. Testing and Evaluation:

The system is tested for recognition accuracy, learning adaptability, usability, and overall performance under real-world conditions.

3.4 Materials and Resources

The development of the system utilizes both hardware and software resources.

Hardware Resources:

A development laptop and Android/iOS mobile devices are used for model training, testing, and real-time deployment.

Software Tools:

Python is used for model development, with TensorFlow or PyTorch for deep learning implementation. OpenCV and MediaPipe support hand tracking and image preprocessing. React Native with Expo Go is used to develop the mobile frontend.

3.5 Anticipated Outcomes

The proposed methodology is expected to produce several positive outcomes.

Personalized Learning Paths:

Learners will experience customized lesson progression based on their individual performance.

Improved Accuracy and Confidence:

Real-time feedback and adaptive repetition will help learners correct mistakes and build confidence.

Reduced Learning Time:

By focusing on mastery-based progression rather than static lessons, learners can achieve proficiency more efficiently.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

i. Sign practice and evaluation - The system allows users to practice Sinhala sign language gestures and evaluates their hand movements using computer vision techniques. User performance is assessed based on gesture accuracy and consistency.
ii. Adaptive lesson control - The learning flow dynamically adjusts based on the learner's performance and progress. Easier or repeated lessons are provided when errors occur, while new signs are introduced after mastery.
iii. Performance tracking - The learning flow dynamically adjusts based on the learner's performance and progress. Easier or repeated lessons are provided when errors occur, while new signs are introduced after mastery.
iv. Real-time feedback - Immediate feedback is provided to inform users whether a gesture is correct or incorrect. Corrective guidance helps learners improve hand positioning and gesture accuracy during practice.
v. Mobile Compatibility – The system must run efficiently on Android and iOS devices using edge-optimized models, ensuring accessibility in everyday contexts.

4.2 Non-Functional Requirements

i. Performance – The system must detect and evaluate hand gestures in real time, providing feedback within one second to maintain an interactive and effective learning experience.
ii. Scalability – The system must support multiple users and expandable lesson content, allowing the addition of new Sinhala letters, numbers, and words without degrading performance.
iii. Usability – The system must provide a simple, intuitive, and visually guided user interface that enables learners to easily understand instructions and perform sign gestures correctly.
iv. Reliability – The system must consistently recognize hand gestures accurately under varying lighting conditions and hand orientations to ensure dependable learning outcomes.
v. Security & Privacy – The system must process camera input locally on the device and store learning progress securely without transmitting or storing sensitive user data externally.

5. GANTT CHART

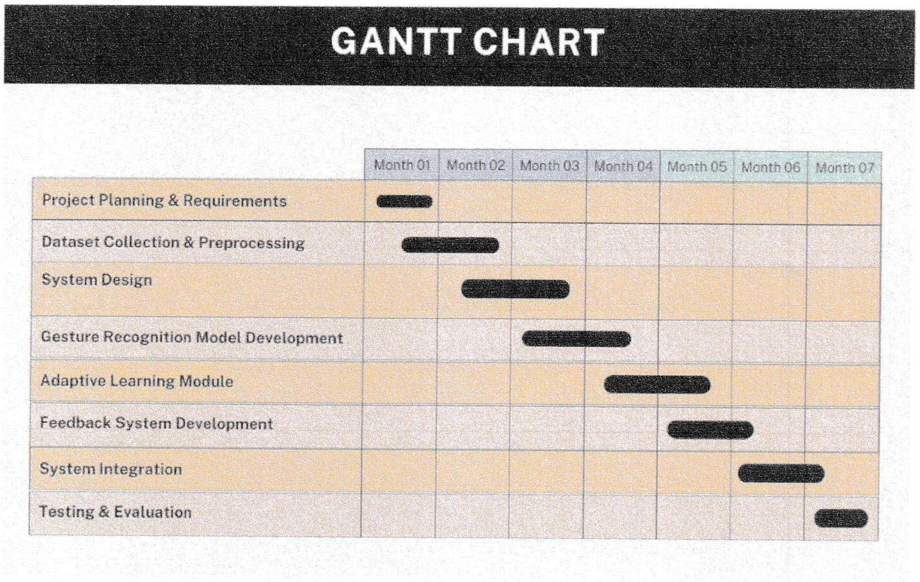

Figure 5-1 Gantt Chart

6. BUDGET AND BUDGET JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Cloud / GPU Usage	25,000	Model training acceleration
Mobile Device for Testing	40,000	To validate real-time performance.
Miscellaneous	5,000	Unexpected expenses.

Table 6-1 Budget Estimation Table

Total Estimated Budget: LKR 70, 000

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